CO_2 and CO adsorption studies on CeO_2-based materials

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Introduction

Ceria (CeO_2) has been evolved as an ubiquitous constituent in catalytic systems, due to its exceptional properties. The facile interplay between Ce^{4+}/Ce^{3+} ions can enhance the amount of oxygen vacancies and the oxygen storage capacity (OSC) of CeO_2, increasing its utilization in different reactions (e.g., exhaust gas treatments, alkane oxidation, and hydrogenation-dehydrogenation reactions). Infrared (IR) spectroscopy is a useful tool to get insights into the surface structure of ceria by studying the number, nature, strength and location of adsorbed species. Especially, the utilization of CO and CO_2 as probe molecules facilitates the bridging between catalytic activity and surface properties in C1 chemistry. Studies have shown that the catalytic and surface properties of CeO_2 are closely related to the exposed crystal facets, with the exposure of the reactive facet depending on the morphology of the material at the nanoscale [1]. However, the assignment of the IR bands to specific species on CeO_2 nanostructures is still challenging as different results have been observed across the literature [2]. In the present work, the adsorption of CO_2 and CO on polycrystalline ceria has been systematically studied by infrared spectroscopy after different pretreatments and under various adsorption conditions. For comparison, the adsorption was investigated on a hydrothermally synthesized material, which consists of nanocubes and is therefore mainly terminated with (100) surfaces.

Results

The infrared spectra obtained after CO_2 adsorption on cerium oxide are generally very complex. Therefore, it is necessary to support the assignment by temperature-programmed experiments. CO_2 adsorption on well-defined nanocubes results mainly in the formation of thermally very stable and highly symmetrical tridentate carbonate species with bands at 1460 cm^{-1} and 1400 cm^{-1} (**Figure 1A**). The assignment is in good agreement with Vayssilov et al. [2]. In addition, hydrogen carbonate is observed, for which the band at 1218 cm^{-1} is characteristic and to which further bands at 3625,

1600 and 1403 cm^{-1} belong. Hydrogen carbonate is also formed on polycrystalline ceria. However, this sample exhibits much greater variability in the symmetry of tridentate carbonates. The greater splitting of the bands from 1570-1280 cm^{-1} indicates that these structures are more distorted. The coadsorption of water partially suppresses the formation of the highly symmetric tridentate carbonates (not shown). Since these symmetric carbonates form at edges and near oxygen defects [2], this observation suggests that water is preferentially adsorbed at these sites. Water also adsorbs at Lewis acid centers, because the band due to molecular adsorbed CO_2 at 2339 cm^{-1} is also suppressed in coadsorption experiments of CO_2 and H_2O.

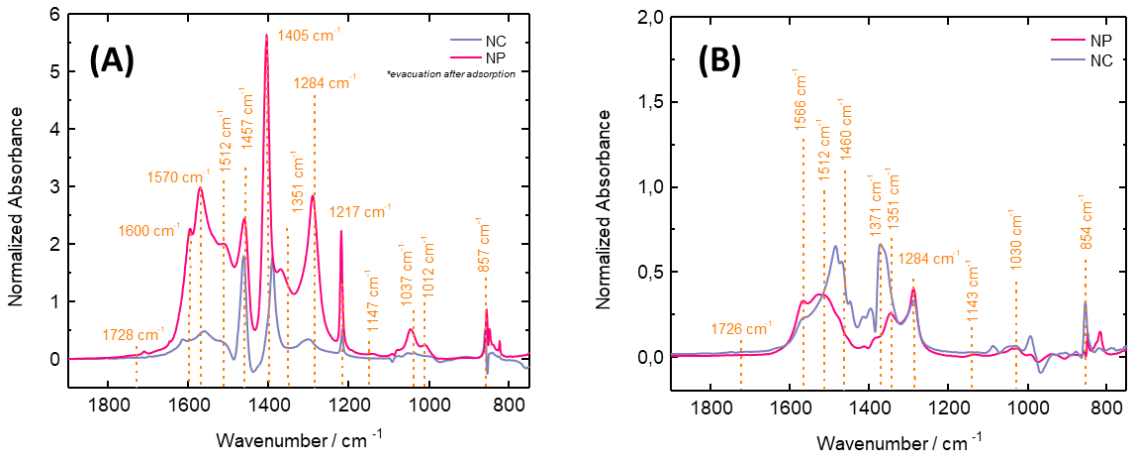

Figure 1: FT-IR spectra of **(A)** 3 mbar CO_2 and **(B)** 52.8 mbar CO at 298 K on ceria nanocubes (grey) and polycrystalline ceria (red), spectra are recorded in vacuum.

Significantly fewer carbonates are formed through CO adsorption (**Figure 1B**). It is interesting to note that Ce-OH groups do not react with CO to form hydrogen carbonates and that, although highly symmetrical tridentate structures are observed on the nanocubes, the latter appear only weakly on polycrystalline ceria. Apparently, there are less oxygen atoms on (100) facets of polycrystalline CeO_2 available that could react with CO to highly symmetrical tridentate carbonates. Formate is also not detected. These adsorption studies are intended as preliminary work to better understand the reactivity of adsorbed species in catalytic processes at the CeO_2 surface.

References

[1] T. Montini, M. Melchionna, M. Monai, P. Fornasiero, Chem. Rev. 116 (2016) 5987–6041.

[2] G.N. Vayssilov, M. Mihaylov, P.St. Petkov, K.I. Hadjiivanov, K.M. Neyman, J. Phys. Chem. C 115 (2011) 23435–23454.